

CHAITHRA BEKAL

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EDUCATION

George Mason University, *MS in Computer Science (Machine Learning)* | Virginia, USA

May 2024

Visvesvaraya Technological University, *B.E in Computer Science* | Karnataka, India

Aug 2020

Courses: Mathematical Foundations | Python Programming | Data Mining | Analysis of Algorithms | Artificial Intelligence | Machine Learning | Advanced Natural Language Processing | Mining Massive Datasets using Spark | Components-based Software Development(AWS, Kubernetes, Docker) | Software Design and Architecture

EXPERIENCE

Inova Systems Corp, *Software Engineer* | Manassas, VA, USA

Aug 2024 - Present

- Collaborate with a cross-functional team of 5 members to identify and analyze potential project opportunities, contributing to the development of 2 project proposals within the first two weeks.
- Lead the technical aspects of proposal writing, detailing software architecture, technology stack, and implementation strategies. Assist in creating project timelines and resource allocation plans.
- Conduct comprehensive market research and competitive analysis, evaluating 15+ similar products to inform proposal strategies and highlight unique selling points.

George Mason University, *Data Analyst Intern* | Fairfax, VA, USA

Feb 2024 - May 2024

- Developed and implemented advanced data cleaning and preprocessing techniques using Excel, Pandas and Python, improving data quality by 40% and reducing analysis time by 25%.
- Created compelling network visualizations using Gephi, Matplotlib and NetworkX, revealing previously unidentified influence patterns and relationships among global military powers.

iD Tech, *Online Instructor in AI and ML* | Remote (USA)

May 2023 - Aug 2023

- Established and executed 10+ AI and ML lesson plans, achieving a 98% success rate for 20+ students in projects.
- Mentored 15+ students in creating innovative applications using OpenAI's API for custom chat systems.
- Innovated hands-on projects in computer vision, natural language processing, and deep learning for object detection.

In Time Tec, *Software Engineer* | India

Jan 2020 - Jun 2022

- Engineered an automation test framework using Selenium with Java, reducing manual testing efforts by 40% and improving overall software quality.
- Implemented CI/CD pipelines with the help of DevOps team for automated testing, achieving a 90% decrease in testing time using Jenkins.
- Conducted root cause analysis for production issues, creating detailed reports and implementing preventive measures to reduce recurring problems by 35%.
- Developed and maintained Java-based applications, resolving critical production issues and implementing new features to meet customer requirements.
- Mentored 6 junior team members in testing methodologies and automation techniques, enhancing team productivity.

SKILLS

Languages and Libraries	Python, SQL, Java, PySpark, Typescript, Numpy, Pandas, Scikit-learn, PyTorch, Matplotlib
Data Science	Data Cleaning and Pre-processing, Data Visualization, Data Analysis
Machine Learning	ML Algorithms, Optimization, Fine-tuning, Metrics, NLP, Neural Networks
Tools	Git, Jira, AWS, Azure ML Studio, Apache Spark

PROJECTS

EconBERTa: Advanced NLP Model for Financial Text Analysis – NLP Paper

Mar 2024 - May 2024

- Developed and fine-tuned EconBERTa language model from Hugging Face, achieving 67% baseline accuracy.
- Conducted comprehensive robustness analysis using Checklist paper resulting in less than 10% failure rate and explored multilingual capabilities.

AI-Powered Fraud Detection: Machine Learning for Job Market Integrity - PySpark

Oct 2023

- Designed a machine learning pipeline using Spark to detect fraudulent job listings achieving 87% accuracy.
- Applied and compared algorithms including Logistic Regression, SVM, Random Forests, and Multilayer Perceptron.

Fake Covid Fact-Checked News Analysis – EDA and Sentiment Analysis

Apr 2023 - May 2023

- Performed in-depth Exploratory Data Analysis on a Fake Covid Fact-Checked News Dataset. Recognized as one of the top 3 best projects in the class for innovative approach and impactful results.
- Mastered sentiment analysis models using NLP techniques across diverse demographics, achieving a 0.93 F1-score.

E-Commerce Intelligence: Sentiment Analysis for Product Improvement

Feb 2023

- Created and implemented a sentiment analysis system for baby product reviews achieving 90% accuracy.
- Leveraged NLTK for text preprocessing and Python, Scikit-learn for model development.